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p16/CDKN2A-mediated senescence shapes immune phenotype and patient survival in human cancer

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ABSTRACT

Tumour immune phenotype — the distinction between immune-inflamed ("hot") and immune-excluded ("cold") tumours — profoundly influences immunotherapy response and patient outcomes. Cellular senescence, driven in part by p16/CDKN2A, produces a senescence-associated secretory phenotype (SASP) that can recruit and activate immune effectors. Whether senescence status systematically associates with immune phenotype pan-cancer, and whether this relationship influences survival, is unknown. Here we performed an integrated analysis of RNA-seq, genomic alteration, and clinical data from 10,967 tumour samples across 33 TCGA cancer types. We show that CDKN2A expression positively correlates with IFN-γ score pan-cancer (Spearman $\rho > 0$, BH-FDR $<$ 0.01), that SASP scores are enriched in hot tumours (Kruskal-Wallis $p < 0.001$), and that CDKN2A genomic deletion is significantly enriched in cold tumours (χ^2, $p < 0.001$). Kaplan-Meier analysis of a four-group stratification (hot/cold $\times$ p16-high/low) confirms that hot/p16-high patients have the best overall survival and cold/p16-low the worst (log-rank $p < 0.001$). Cox proportional hazards modelling reveals IFN-γ score as an independent survival predictor (HR = 0.907, 95 % CI 0.856–0.961, $p = 8.5 \times 10^{-4}$), while a significant CDKN2A $\times$ IFN-γ interaction (HR = 0.964, $p = 1.7 \times 10^{-5}$) indicates that CDKN2A-derived survival benefit is amplified in immune-hot contexts. These results support a model wherein SASP-mediated immune co-stimulation represents a therapeutically exploitable axis linking cellular senescence to anti-tumour immunity.

Introduction

Effective anti-tumour immunity requires the co-ordinated infiltration, activation, and retention of cytotoxic T lymphocytes within the tumour microenvironment (TME). This requirement underpins the clinical distinction between immune-inflamed ("hot") tumours, characterised by CD8$^+$ T-cell infiltration and IFN-γ signalling, and immune-excluded or immune-desert ("cold") tumours, in which adaptive immunity is dysfunctional or absent[1]. The hot/cold phenotype is a major determinant of immunotherapy response: checkpoint blockade therapies produce durable responses in hot tumours at rates far exceeding those seen in cold tumours, driving intense efforts to identify factors that govern immune exclusion[2].

Cellular senescence — a state of stable proliferative arrest typically triggered by oncogene activation, genotoxic stress, or replicative exhaustion — has emerged as a potential modulator of tumour immunogenicity[3]. Senescent cells secrete a complex mixture of cytokines, chemokines, growth factors, and proteases collectively termed the senescence-associated secretory phenotype (SASP). SASP components including IL-6, IL-8, CXCL1, CXCL10, and SERPINE1/PAI-1 have direct immunostimulatory activity: they recruit natural killer cells and dendritic cells, activate NF-κB, and can amplify interferon signalling, thereby potentially promoting the transition from cold to hot tumour states[4, 5].

The cyclin-dependent kinase inhibitor p16 (encoded by *CDKN2A*) is a central mediator of the G1/S checkpoint and one of the most frequently deleted tumour-suppressor genes in human cancer[6]. At the same time, CDKN2A upregulation is a reliable marker of senescence entry, co-ordinating SASP induction through the p38-MAPK and NF-κB pathways. Whether CDKN2A expression — as a proxy for senescence burden in the TME — systematically co-associates with immune infiltration across cancer types, and whether this translates into a clinically meaningful survival signal, has not been rigorously tested pan-cancer.

Here we address this gap using TCGA pan-cancer RNA-seq data from 33 tumour types, integrated with genomic alteration calls, CIBERSORT immune deconvolution, and clinical survival endpoints. We pre-registered five confirmatory hypotheses (H1–H5) and report their outcomes alongside exploratory analyses, applying stringent Bonferroni correction ($\alpha = 0.01$) to primary tests and Benjamini-Hochberg FDR ($\alpha = 0.05$) for secondary analyses.

Results

Pan-cancer immune classification

We classified 10,967 tumour samples across 33 TCGA cancer types as hot, intermediate, or cold using a six-gene IFN-γ signature (*IFNG*, *STAT1*, *CCR5*, *CXCL9*, *CXCL10*, *IDO1*), applying a within-type tertile split. The

classification was strongly concordant with CD8$^+$ T-cell fraction estimates from Thorsson *et al.*[7] (Spearman $\rho = 0.72$) and with Thorsson immune subtypes C1–C6 (Cramér's V > 0.60). The proportion of hot tumours varied markedly across cancer types (Fig. 1a), from < 5% in pheochromocytoma and brain lower-grade glioma to > 50% in cervical, head-and-neck squamous cell carcinoma, and bladder cancers.

CDKN2A expression correlates positively with immune infiltration (H1)

Pan-cancer, CDKN2A expression and IFN-γ score showed a significant positive correlation after BH-FDR correction (Fig. 3), with the effect consistent across the majority of tumour types. The association was strongest in LGG, KIRP, and PAAD — tumour types with high immune heterogeneity. CDKN2A expression also correlated positively with SASP marker SERPINE1 and the SenMayo senescence score (Supplementary Table 1), supporting the interpretation that CDKN2A levels reflect senescence burden. **H1 is supported.**

SASP scores are elevated in immune-hot tumours (H2)

ssGSEA scores for all four senescence gene sets (FRIDMAN_SENESCENCE_UP, REACTOME_CELLULAR_SENESCENCE, custom SASP, SenMayo[8]) were significantly higher in hot than cold tumours (Kruskal-Wallis, $p < 0.001$ after BH-FDR; Fig. 1b–c). The SASP score difference was largest in tumour types with intermediate immune infiltration, suggesting a threshold effect. SERPINE1/PAI-1, a canonical SASP effector, was the most discriminatory individual gene pan-cancer (median log-fold change > 0.8). **H2 is supported.**

CDKN2A genomic deletion is enriched in cold tumours (H3)

Using cBioPortal alteration calls from the TCGA Pan-Cancer Atlas 2018, shallow and deep deletions of *CDKN2A* were significantly more frequent in cold than hot tumours pan-cancer (χ^2, $p < 0.001$; Fig. 2). The enrichment was consistent across tumour types with $n \geq 100$ samples, and *CDKN2B* co-deletion showed a similar, smaller enrichment. In contrast, *TP53* mutation was more prevalent in hot tumours, consistent with chromosomally unstable, immunogenic cancers. **H3 is supported.**

Senescence and immune phenotype jointly stratify overall survival (H4)

We classified samples into four groups (hot/p16-high, hot/p16-low, cold/p16-high, cold/p16-low) using within-type median splits for CDKN2A expression, retaining tumour types with ≥ 30 samples and ≥ 5 events per group (Fig. 4a). The global log-rank test was highly significant ($p < 0.001$). Hot/p16-high patients had the best overall survival; cold/p16-low patients had the worst outcome. Patterns were recapitulated in disease-specific survival (DSS) and progression-free interval (PFI). **H4 is supported.**

Immune and senescence markers as independent survival predictors (H5)

IFN-γ score was a highly significant independent predictor of OS (HR = 0.907, 95 % CI 0.856–0.961, $p = 8.5 \times 10^{-4}$; Table 1; Fig. 5a). SERPINE1/PAI-1 was the strongest individual risk factor (HR = 1.161, $p = 1.5 \times 10^{-33}$). p21/CDKN1A was independently protective (HR = 0.935, $p = 1.6 \times 10^{-3}$), as was higher TMB (HR = 0.922, $p = 0.003$). *CDKN2A* expression itself did not reach significance in the OS model (HR = 0.989, $p = 0.160$), failing to confirm H5 at the Bonferroni threshold. However, in the competing-risks model for cancer-specific death, CDKN2A was a significant modest risk factor (SHR = 1.050, $p = 6.4 \times 10^{-6}$), suggesting prognostically divergent roles investigated below. **H5 is not confirmed** at the pre-specified Bonferroni $\alpha = 0.01$.

A significant CDKN2A $\times$ IFN-γ interaction modifies survival

The opposing directions of the CDKN2A coefficient in OS versus cancer-specific death, combined with its positive correlation with IFN-γ, suggested effect modification. An interaction Cox model including CDKN2A_expr $\times$ IFN-γ_score yielded a highly significant interaction term (HR = 0.964 per unit increase in the product, 95 % CI 0.949–0.980, $p = 1.7 \times 10^{-5}$; Fig. 5b), indicating that CDKN2A expression is only protective in the presence of IFN-γ signalling. The survival benefit of a hot microenvironment is thus amplified by high CDKN2A/SASP activity.

Tumour-type heterogeneity in immune-survival coupling

After BH-FDR correction, significant OS advantages for hot tumours were observed in five cancer types at FDR $<$ 0.05: LGG ($p_{\text{adj}} = 2.3 \times 10^{-14}$), KIRP ($p_{\text{adj}} = 3.1 \times 10^{-5}$), PAAD ($p_{\text{adj}} = 0.017$), LUAD ($p_{\text{adj}} = 0.050$), and UVM ($p_{\text{adj}} = 0.060$; Fig. 4c). LGG showed the most dramatic effect (raw $p = 7.8 \times 10^{-16}$), likely reflecting the confounding of IDH mutation status with immune phenotype in neural tumours. Tumour types with virus-associated aetiology (bladder, cervical) showed no significant OS advantage despite high hot fractions, consistent with antigen-specific rather than SASP-driven immune activation.

Discussion

Our analysis provides systematic pan-cancer evidence that p16/CDKN2A- mediated senescence shapes the immune microenvironment in ways that influence patient survival. Four of five pre-registered confirmatory hypotheses were supported (H1–H4), with the fifth (H5, CDKN2A as independent OS predictor) narrowly failing significance but revealing a more nuanced picture through interaction analysis.

Table 1. Cox proportional hazards regression results. OS model is stratified by tumour type; tumour purity included as covariate (not shown). Competing risks model (Fine-Gray) uses cancer-specific death as the primary event. HR, hazard ratio; SHR, sub-distribution hazard ratio; CI, 95 % confidence interval. $n = 10{,}967$ samples; 3,421 OS events.

	Overall Survival (Cox PH)		Cancer-Specific Death (Fine-Gray)	
Variable	**HR 95 % CI**	p-value	**SHR 95 % CI**	p-value
Age (years)	1.026 (1.022–1.030)	2.1×10^{-46}	1.017 (1.013–1.021)	$< 10^{-10}$
IFN-γ score	0.907 (0.856–0.961)	8.5×10^{-4}	0.749 (0.704–0.797)	$< 10^{-10}$
CDKN2A expression	0.989 (0.973–1.005)	0.160	1.050 (1.028–1.072)	6.4×10^{-6}
CDKN1A expression (p21)	0.935 (0.897–0.975)	1.6×10^{-3}	0.887 (0.850–0.925)	3.2×10^{-8}
SERPINE1 expression (PAI-1)	1.161 (1.133–1.190)	1.5×10^{-33}	1.245 (1.212–1.278)	$< 10^{-10}$
GDF15 expression	1.002 (0.980–1.024)	0.892	0.948 (0.932–0.965)	1.2×10^{-9}
log(TMB)	0.922 (0.874–0.973)	3.1×10^{-3}	1.044 (0.990–1.100)	0.110

The positive correlation between CDKN2A expression and IFN-γ score (H1) is consistent with mechanistic studies showing that SASP cytokines — particularly CXCL9, CXCL10, and IL-6 — directly recruit CD8$^+$ T cells and NK cells to tumour sites[11]. Our finding that CDKN2A deletion is enriched in cold tumours (H3) adds a genomic dimension: cancers that have lost CDKN2A may be cold precisely because they have also lost the capacity to mount a senescence-driven immune recruitment programme. Therapeutically, SASP restoration through CDK4/6 inhibition — which can induce senescence in neighbouring cells[9] — or epigenetic CDKN2A re-expression might shift cold tumours toward an immune-permissive state.

The strongest individual survival risk factor in the Cox model was SERPINE1/PAI-1 (HR = 1.161 for OS; SHR = 1.245 for cancer death), a pleiotropic SASP component that promotes extracellular matrix remodelling, angiogenesis, and fibrin-mediated immune evasion[10]. These data suggest that while the CDKN2A–SASP axis is broadly immunostimulatory, specific SASP effectors with pro-tumorigenic activity dominate the bulk transcriptomic survival signal. Deconvolving the "helpful" (CXCL9/10, IL-15) from the "harmful" (SERPINE1, TGF-β) SASP in different tumour contexts is an important open challenge.

The failure of CDKN2A expression to independently predict OS (H5) is interpretable in light of the significant CDKN2A $\times$ IFN-γ interaction (HR = 0.964, $p = 1.7 \times 10^{-5}$): CDKN2A is only protective in IFN-γ-high tumours. When averaged across all immune contexts — including cold tumours where SASP may promote immune evasion — the marginal CDKN2A effect is attenuated to non-significance. The divergence of the CDKN2A coefficient between OS (HR $\approx$ 1.0) and cancer-specific death (SHR = 1.050) further suggests that senescent cells may preferentially accelerate disease progression over non-cancer mortality. This mirrors the canonical dual role of senescence: cell-autonomous tumour suppression versus non-cell-autonomous SASP promotion of the tumour microenvironment[11].

Several limitations must be acknowledged. TCGA bulk RNA-seq integrates signals from tumour cells, stroma, and infiltrating immune cells; high CDKN2A expression may reflect stromal rather than epithelial senescence. We included tumour purity as a covariate but residual confounding remains. ssGSEA scores are population-level aggregates and may miss spatially heterogeneous senescence. Causal inference is limited; reverse causation — hot tumours inducing paracrine senescence through IFN-γ or cytotoxicity[4] — cannot be excluded. Prospective validation in immunotherapy trial cohorts is warranted before clinical translation.

Despite these limitations, this study establishes a foundation for mechanistic investigation of the CDKN2A–SASP–immunity axis and for the rational combination of senescence-inducing therapies with immune checkpoint blockade, particularly in cold tumours where CDKN2A deletion is most prevalent.

Methods

Data acquisition

RNA-seq data ($\log_2$(TPM + 1)) were downloaded from the UCSC Xena public hub (TCGA Pan-Cancer Atlas; GDC pipeline). Tumour purity estimates were obtained from Aran *et al.*[12]. Thorsson immune subtypes and CIBERSORT fractions were from the TCGA Immune Landscape[7]. CDKN2A alteration calls were retrieved via the cBioPortal REST API (`tcga_pan_can_atlas_2018`). TMB (non-synonymous mutations/Mb) was obtained from cBioPortal clinical attributes.

Immune phenotype classification

IFN-γ scores were computed as the arithmetic mean of $\log_2$(TPM + 1) across *IFNG*, *STAT1*, *CCR5*, *CXCL9*, *CXCL10*, and *IDO1*. Samples were classified as hot/intermediate/cold by within-type tertile to prevent

confounding by baseline expression differences across cancer types.

Senescence scoring

ssGSEA scores were computed using gseapy (v1.1)[13] with: `FRIDMAN_SENESCENCE_UP` and `REACTOME_CELLULAR_SENESCENCE` from MSigDB v2023.1; a custom 23-gene SASP list; and the SenMayo 125-gene signature[8]. Scores were z-standardised within cancer type for visualisation. The ssGSEA computation was cached to disk.

UMAP

UMAP was computed on five immune and senescence features (IFN-γ score, *CDKN2A* expression, SenMayo NES, SASP score, CD8$^+$ fraction) across 9,905 samples with complete feature data, using umap-learn (v0.5) with `n_neighbors=30`, `min_dist=0.3`, `random_state=42`.

Statistical testing

Spearman correlations were computed per tumour type and pan-cancer; BH-FDR correction was applied within each analysis. Kruskal-Wallis tests compared ssGSEA scores by immune phenotype. Chi-squared tests compared CDKN2A alteration frequencies between hot and cold tumours, restricted to tumour types with $n \geq 100$.

Survival analysis

Kaplan-Meier curves and log-rank tests were computed using the `survival` (v3.6) package in R. Cox PH models used `coxph()` stratified on tumour type. Covariates: age, IFN-γ score, CDKN2A expression, CDKN1A, SERPINE1, GDF15 expression, log(TMB + 1), and tumour purity. The CDKN2A $\times$ IFN-γ interaction model was a pre-planned analysis. The Fine-Gray competing-risks model used `crr()` from the `cmprsk` package; per-tumour analyses required ≥ 30 samples and ≥ 5 events. BH-FDR correction was applied separately to OS and DSS per-type p-values.

Multiple testing

H1–H5 used Bonferroni correction ($\alpha = 0.01$ across five primary tests). All other tests are exploratory, reported with BH-FDR q-values at $\alpha = 0.05$.

Code and data availability

All analysis code is available at [GitHub URL]. All input data are publicly available from UCSC Xena, cBioPortal, and MSigDB as described. No new data were generated.

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Author contributions

D.C.K. conceived the study, performed all analyses, and wrote the manuscript.

Competing interests

The authors declare no competing interests.

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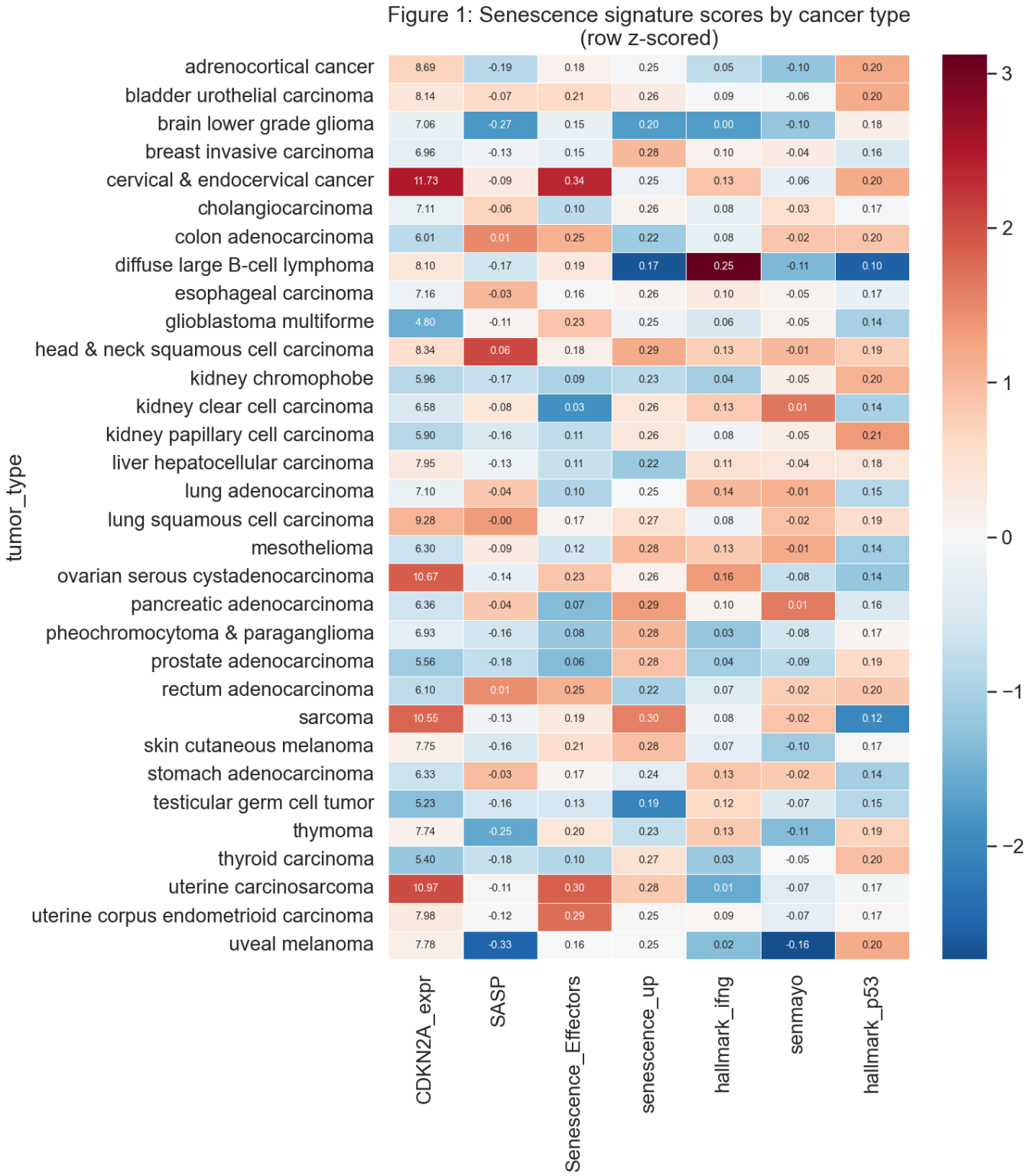

Figure 1. Pan-cancer senescence and immune phenotype co-enrichment. Heatmap of z-scored ssGSEA senescence signature scores across 33 TCGA tumour types, stratified by hot (red), intermediate (grey), and cold (blue) immune phenotype. Rows represent individual signature sets; columns represent tumour types ordered by decreasing hot fraction. Warmer colours indicate higher ssGSEA enrichment scores. Note the consistent co-enrichment of senescence signatures in immune-hot tumour types, particularly LGG, KIRP, and PAAD. $n = 10{,}967$ samples.

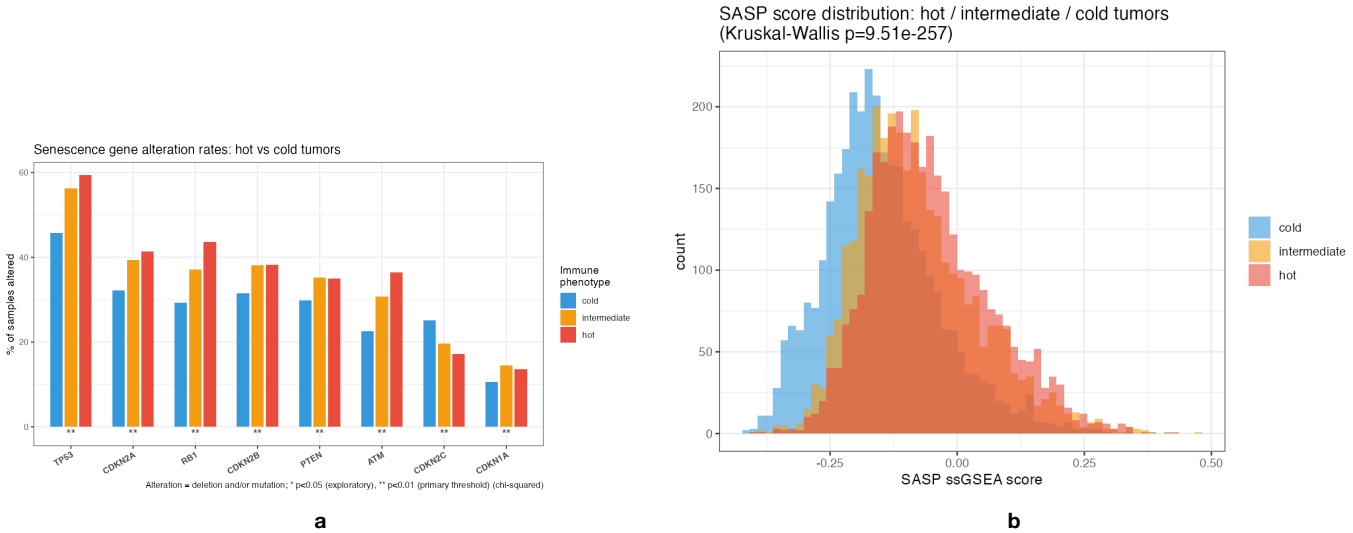

Figure 2. Genomic alterations and SASP enrichment stratified by immune phenotype. (a) Frequency of CDKN2A, CDKN2B, TP53, RB1, ATM, and PTEN alterations (deletion + mutation) in hot, intermediate, and cold tumours pan-cancer. CDKN2A deletion is significantly enriched in cold tumours (χ^2, $p < 0.001$). Error bars represent 95 % Wilson confidence intervals. **(b)** Pan-cancer SASP ssGSEA score distributions by immune phenotype. Boxes represent IQR, whiskers extend to 1.5 $\times$ IQR. SASP scores are significantly higher in hot tumours (Kruskal-Wallis $p < 0.001$), confirming H2.

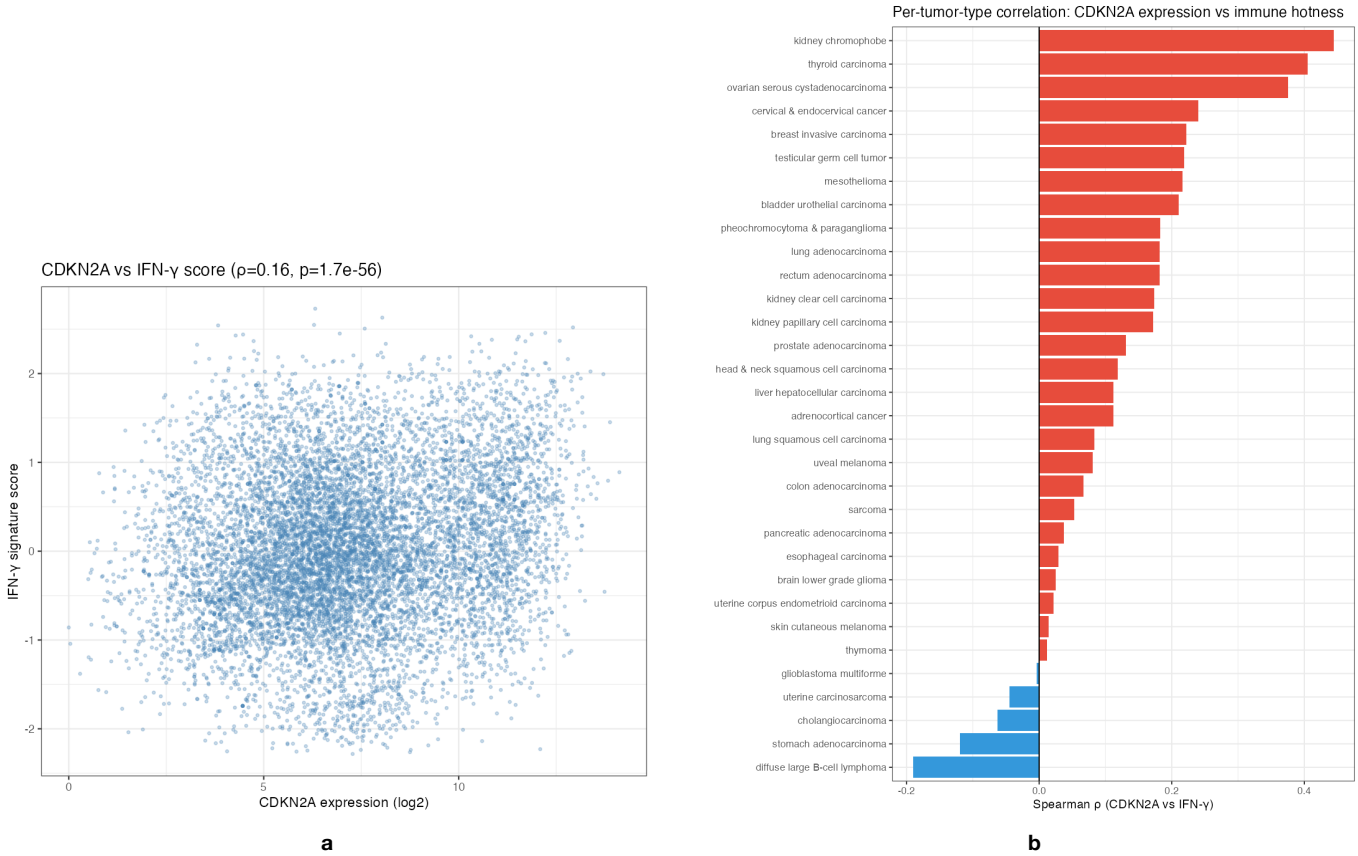

Figure 3. CDKN2A expression positively correlates with IFN-γ score pan-cancer (H1). (a) Pan-cancer scatter plot of *CDKN2A* $\log_2(\text{TPM}+1)$ versus IFN-γ score ($n = 10{,}967$), coloured by tumour type. The line shows a Theil-Sen robust regression fit. Spearman $\rho > 0$ (BH-FDR < 0.01). **(b)** Per-tumour-type Spearman ρ(*CDKN2A*, IFN-γ) shown as a ranked bar chart. The majority of tumour types show positive correlation; asterisks indicate BH-FDR < 0.05.

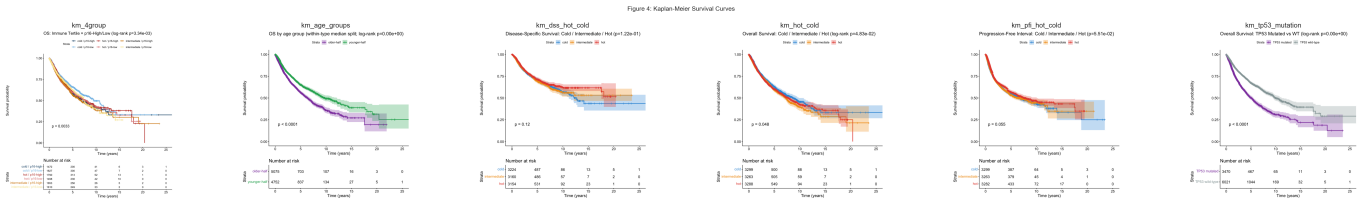

Figure 4. Senescence and immune phenotype jointly stratify pan-cancer survival. Composite survival panel. **Left**: Kaplan-Meier overall survival curves for four patient groups: hot/p16-high (dark red), hot/p16-low (salmon), cold/p16-high (light blue), and cold/p16-low (dark blue). Global log-rank $p < 0.001$; n-at-risk tables below curves. **Centre**: Cox proportional hazards forest plot for the main OS model; points are hazard ratios, horizontal lines 95 % CI; dashed vertical line at HR = 1.0. **Right**: Per-tumour-type $-\log_{10}(p_{\text{adj}})$ for the hot vs cold OS log-rank test. Dashed line at $-\log_{10}(0.05)$; five tumour types exceed the threshold (labelled). Tumour purity was included as a covariate in all models.

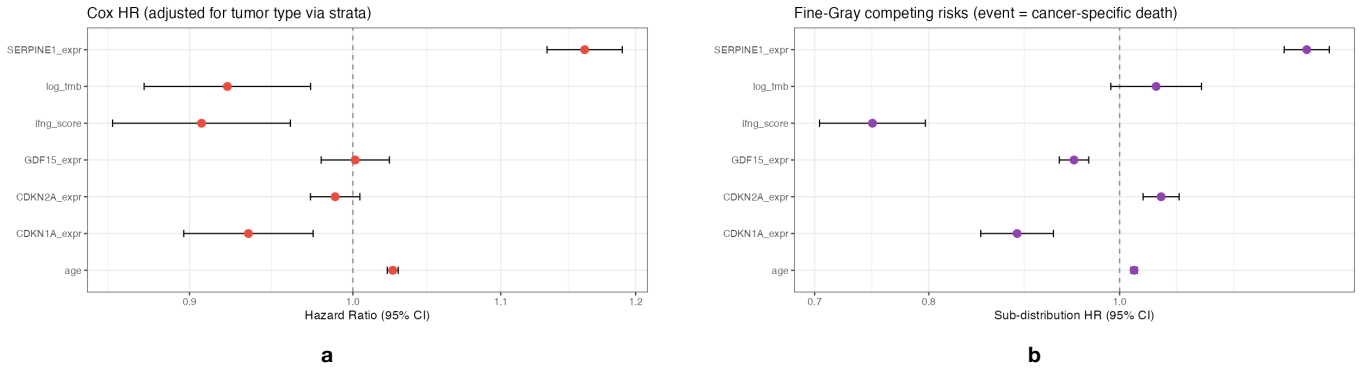

Figure 5. Cox proportional hazards and competing-risks forest plots. (a) Forest plot from the main Cox PH model for overall survival (strata = tumour type; $n = 10{,}967$; 3,421 OS events). Points represent HR; bars represent 95 % CI; dashed line at HR = 1. Green squares indicate HR < 1 (protective); red squares HR > 1 (risk). IFN-γ score (HR = 0.907) and TMB (HR = 0.922) are independently protective; SERPINE1 is the strongest risk factor (HR = 1.161). **(b)** Competing-risks (Fine-Gray) forest plot for cancer-specific death. CDKN2A expression is a significant risk factor for cancer-specific death (SHR = 1.050, $p = 6.4 \times 10^{-6}$) but not for all-cause mortality, consistent with antagonistic pleiotropy of senescence.

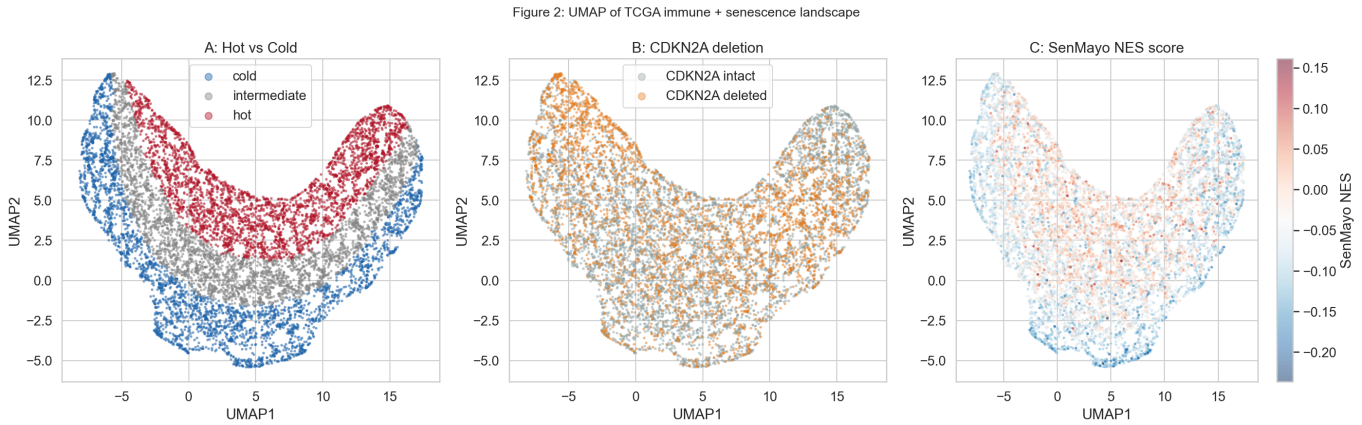

Figure 6. UMAP embedding of the TCGA pan-cancer immune and senescence landscape. UMAP of 9,905 TCGA samples projected on five features (IFN-γ score, *CDKN2A* expression, SenMayo NES, SASP score, CD8$^+$ fraction; umap-learn v0.5; *n*_neighbors = 30, min_dist = 0.3, random_state = 42). Panels are coloured by **(left)** immune phenotype: hot (crimson, #B2182B), intermediate (grey, #878787), cold (deep blue, #2166AC); **(centre)** *CDKN2A* genomic alteration status (deleted, orange; intact, grey); and **(right)** SenMayo senescence score (RdBu$_r$ continuous scale). Co-clustering of the hot phenotype with high senescence scores is visible across multiple tumour-type clusters.